

DesignOS

From an AI App to a Native Knowledge Operating System

A source-grounded case study of the architecture, evolution and evaluation behind DesignOS - spanning 41 specialized engines, an 18-format ingestion layer, and the V30 Native Knowledge Center that turns fragmented product work into a governed, reusable knowledge system.

41

Specialized engines

18

Input formats

32/32

Verification suites passed

99.8

Overall quality score /100





”

Knowledge stops being a passive file collection and becomes a native operating layer.

- V30 Master Walkthrough, §07

CASE STUDY BOOKLET
11 PAGES · A4 LANDSCAPE

DESIGNED BY
Prashant Sisodhiya

01

OVERVIEW

A quick introduction to DesignOS, its purpose and the knowledge-operating-system vision.

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PROBLEM & ASSUMPTIONS

Why fragmented product work forced the architecture, and the four bets it makes.

PAGE 04

03

PROCESS

Evolution across 25 verified stages, V18 to V30 - release, evaluation, knowledge, orchestration.

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ARCHITECTURE

The V30 master pipeline - 16 stages from raw idea to certified, multi-format output.

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KNOWLEDGE SYSTEM

Inside the Native Knowledge Center - ingestion, search latency, graph and Project Brain.

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STRATEGY

The Golden Knowledge Corpus direction - governed domains, atomicity, evidence and lifecycle.

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ROADMAP

Eight strategic stages from architecture completion to validated cross-project knowledge.

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RESULTS

Verified results - 32/32 suites passed, 99.8/100 quality score, full system scale.

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09

CONCLUSION & TAKEAWAYS

Final verdict, four key takeaways and the next milestone for the corpus.

PAGE 11

01

01 • OVERVIEW

CASE STUDY
SECTION

Governed Knowledge.
Coordinated Engines.
Certified Output.

DesignOS is a source-grounded AI knowledge operating system that turns fragmented product work - discovery, research, UX, UI, architecture and publication - into a single governed corpus that 41 specialized engines read and write.

- Native Knowledge Center
 - Golden Corpus Governance
 - Quality Certification Gate
- 41-Engine Orchestration
 - Multi-format Publication
 - Cross-Project Rollout



PROJECT	MY ROLE	COVERAGE	PLATFORM	TEAM	TOOLS
DesignOS - AI-native product intelligence & knowledge platform	Product Designer • UX Architect Case Study Author	V18 → V30 25 verified stages	Web app • Figma plugin Multi-format export	Product • AI/Backend Frontend • QA	Figma • FigJam • Notion Antigravity IDE

"Architecture is the moat – the corpus is the compounding advantage."
— Prashant Sisodhiya, Product Designer • UI/UX • Creative Technologist

PS

PRASHANT SISODHIYA
PRODUCT DESIGNER • UI/UX • AI SYSTEMS

02

02 • PROBLEM & ASSUMPTIONS

CASE STUDY
SECTION

Why DesignOS needed to exist

The problem statement below is stated directly in the source documentation; the assumptions are the underlying design bets DesignOS makes to address it.

THE PROBLEM • Product work is fragmented

Complex product work is fragmented across discovery, research, strategy, requirements, UX, UI, design systems, technical architecture, documentation, publication and evaluation. DesignOS progressively addresses these problems through specialized but coordinated layers.

10 FRAGMENTED DOMAINS (as documented)

Discovery

Strategy

UX

Design systems

Documentation

Research

Requirements

UI

Tech architecture

Publication & evaluation

CONSEQUENCES OF FRAGMENTATION

- Context loss
- Repeated work
- Inconsistent terminology
- Contradictory assumptions
- Weak validation
- Poor institutional memory

DESIGN ASSUMPTIONS • Bets the architecture makes

A1

§13

Architecture beats model choice

The strongest differentiation is architectural, not model-specific - orchestration, planning and evaluation matter more than which model generates the output.

A2

§14

Sophistication ≠ quality

A sophisticated architecture does not automatically produce excellent output - quality depends on source knowledge, retrieval accuracy and human feedback.

A3

§17

AI output isn't trusted by default

AI-generated content must not automatically become Golden Knowledge - it requires evaluation, evidence validation and human review.

A4

§24

Reported metrics ≠ proven metrics

Implementation-reported results (31-suite pass, A+ certification) must remain live production measurements, not static claims.

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PRASHANT SISODHIYA

PRODUCT DESIGNER • UI/UX • AI SYSTEMS

RELEASE & BUILD

P2 IDE — Intelligent Development Execution

EVALUATE & REVIEW

P5 HAIRB+DCMOI — Human + AI Review Board

KNOWLEDGE MEMORY

P7B KGI+RIT — Knowledge Graph Intelligence

ORCHESTRATION & PUBLICATION

V19 PAE+AEBL — Publication Architecture Engine

V21 PEEE — Publication Experience Evaluation

V23 ERE+AAP — Editorial Reasoning Engine

COGNITION & PLANNING

V26 ACL — AI Cognition Layer

V26.2 DWP+REE — Workflow & Resource Planning

KNOWLEDGE GOVERNANCE

V27.1 KPGP — Knowledge Production & Governance

V29 QEP — Quality Excellence Program

18

15

5

41

engines certified on NKC

Raw Human Idea → NKC → QEP → PRCP → orchestration & authoring engines → multi-format renderers → certified output.
(Documented master pipeline)

- 1 Raw Human Idea**
18 input formats · text, paste, upload

2 NKC
Native Knowledge Center - detect, classify, chunk, embed, graph

3 QEP
Quality Excellence Master Framework - Grade A+

4 PRCP
Production Readiness & Certification Gate - 99.8/100

5 KPGP / KFEL
Knowledge Governance & Factory

6 DWP / REE
Workflow & Resource Planning

7 PDIS
Project Discovery & Intake

8 ACL
AI Cognition Layer

9 ATPE
Adaptive Task Planning & Execution

10 UIOE
Unified Intelligence Orchestration

11 ERE / AAP
Editorial Reasoning Engine

12 DAES
Document Authoring Excellence

13 PAE / AEBL
Publication Architecture Engine

14 APCE / PDT
Adaptive Page Composition

15 PEEE
Publication Experience Evaluation

16 PECVP
Continuous Validation Platform

PDF DOCX HTML Markdown PPTX Figma JSON → NKC

CERTIFICATE · DESIGNOS NKC CERTIFIED & ACTIVE — Grade A+, 99.8/100

PS **PRASHANT SISODHIYA**
PRODUCT DESIGNER - UI/UX - AI SYSTEMS

0505 • KNOWLEDGE SYSTEM

CASE STUDY SECTION

Inside the Native Knowledge Center

Real system data from the V30 engine: ingestion, search latency, graph structure, and live Project Brain sync counts.

INGESTION PATH §07 • Knowledge Processing

1. Detect Format
2. Classify
3. Extract Structure
4. Chunk
5. Embed
6. Create Relationships
7. Sync Project Brain
8. Index Search
9. Evaluate
10. Version
11. Distribute

INPUT COVERAGE

18 documented input formats

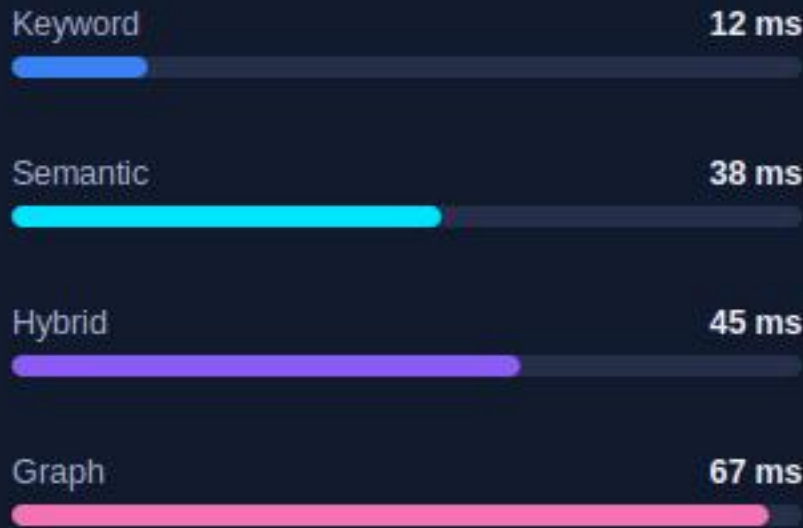
- Markdown
- Plain Text
- YAML
- JSON
- Mermaid
- PDF
- DOCX
- PPTX
- Image
- ZipProject
- FigmaExport
- Code
- ResearchReport
- CaseStudy
- PRD
- DesignSystem
- TechnicalDoc
- Portfolio

CLASSIFICATION

15 classification categories

- Project
- PRD
- CaseStudy
- Research
- Blueprint
- Prompt
- Diagram
- API
- Database
- DesignSystem
- Prototype
- Architecture
- Presentation
- Portfolio
- TechDocs

ENTERPRISE SEARCH — LIVE LATENCY



4 search modes - Keyword, Semantic, Hybrid, Graph - run over the same governed knowledge graph, letting retrieval move from plain document lookup to relationship-aware reasoning.

KNOWLEDGE GRAPH

16 edges across 8 nodes • 5 types

- | | |
|------------|---|
| References | 8 |
| ChildOf | 8 |
| DependsOn | 1 |
| Generates | 1 |
| VersionOf | 1 |

PROJECT BRAIN — LIVE SYNC

- | | |
|--------------|----|
| Decisions | 12 |
| Requirements | 8 |
| Research | 15 |
| Architecture | 1 |
| Roadmap | 6 |

Synchronizes structured project memory so every downstream engine reasons with full context instead of an isolated prompt.

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PRASHANT SISODHIYA

PRODUCT DESIGNER • UI/UX • AI SYSTEMS

0606 • STRATEGY

CASE STUDY SECTION

The Golden Knowledge Corpus

V31 direction: populate NKC with governed, evidence-aware, reusable knowledge. §15-18

GOVERNED DOMAINS §15



KNOWLEDGE ATOMICITY §16 • 15 reusable atomic object types



Large documents are complemented by atomic objects, enabling precise retrieval and relationship-aware reasoning rather than whole-document search.

EVIDENCE & TRUST §17 • 8 confidence levels, low → high



low trust

high trust →

AI-generated output must not automatically become Golden Knowledge - promotion requires evaluation, evidence validation, consistency checks and human review.

KNOWLEDGE LIFECYCLE §18 • 11-stage governed lifecycle, RAW → ARCHIVED

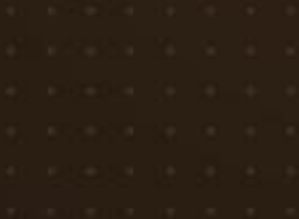


Version history remains immutable enough to support rollback and full auditability at every stage.

0707 • ROADMAP

CASE STUDY SECTION

Strategic roadmap — 8 stages §26



The path from architecture completion to validated, reusable cross-project knowledge.

HARDEN THE CURRENT PLATFORM • stages 1–6

- 1Complete architecture
All 41 engines wired, verified, certified.
- 2Populate the Golden Corpus
Governed knowledge across 21 domains.
- 3Build retrieval evaluation
Precision, recall, grounded-answer rate.
- 4Benchmark generation
Original benchmark families per document type.
- 5Validate visual rendering
Layout defect rate, export fidelity.
- 6Human validation
Structured review before promotion to GOLDEN.

MULTI-PROJECT EXPANSION • stages 7–8

- 7Expand to CureIQ, FutureMind, TradeX, CosmosX, SecureX, PathWise X, DesignOS
Cross-project rollout.
- 8Extract validated cross-project knowledge
Only proven knowledge graduates.

Stages 1–6 harden the platform on its current codebase. Stages 7–8 are the multi-project expansion.

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PRASHANT SISODHIYA
PRODUCT DESIGNER • UI/UX • AI SYSTEMS

0909 • CONCLUSION & TAKEAWAYS

CASE STUDY
SECTION



DesignOS has an architecturally mature, highly orchestrated engine layer - the next leap in quality now depends on populating and governing the Golden Knowledge Corpus, not on adding more engines.

FINAL VERDICT

Architecture Highly mature	Orchestration Strong	Knowledge infra Strong foundation	Publication Strong	Governance Strong
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KEY TAKEAWAYS

- 01

Architecture is the moat

Orchestration, planning and evaluation differentiate more than model choice alone.
- 02

Knowledge must be governed, not assumed

AI output earns trust through evaluation and evidence - it isn't Golden by default.
- 03

Verification must be continuous

31-suite certification stays meaningful only as a live, repeatable measurement, not a static claim.
- 04

Quality now depends on the corpus

The next leap comes from populating and governing the Golden Knowledge Corpus, not from adding engines.

NEXT MILESTONE • Golden Knowledge Corpus



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PRASHANT SISODHIYA
PRODUCT DESIGNER • UI/UX • AI SYSTEMS

Smarter Architecture. Deeper Intelligence.